International AS · Edexcel · Chemistry

22 mins · 17 questions

Multiple Choice Questions

Redox Chemistry & Acid-Base Titrations

Oxidation Numbers - Introduction / Oxidation & Reduction / Disproportionation Reactions / Ionic Equations / Acid-Base Titrations with Indicators

Easy (3 questions)	/3
Medium (13 questions)	/16
Hard (1 question)	/3
Total Marks	**/22**

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Easy Questions

1 What is the oxidation number of chromium in $Na_2Cr_2O_7$?

A. +1

B. +2

C. +3

D. +6

(1 mark)

2 In an oxide of nitrogen, the oxidation number of nitrogen is +4.

Which is the formula of the oxide?

A. N_2O

B. N_2O_3

C. N_2O_4

D. N_2O_5

(1 mark)

3 What is the oxidation number of phosphorus in the phosphate ion, PO_4^{3-}?

A. –3

B. +3

C. +5

D. +7

(1 mark)

Medium Questions

1 Hydrogen peroxide, H_2O_2, breaks down into water and oxygen.

In terms of oxidation and reduction, how do hydrogen and oxygen change in this reaction?

		Hydrogen	**Oxygen**
☐	**A**	oxidised	reduced
☐	**B**	oxidised and reduced	unchanged
☐	**C**	reduced	oxidised
☐	**D**	unchanged	oxidised and reduced

(1 mark)

2 Iodate(V) ions, IO_3^-, oxidise dithionate ions, $S_2O_6^{2-}$, according to the equation

$$\mathbf{x}\ IO_3^- + \mathbf{y}\ S_2O_6^{2-} + 4H_2O \rightarrow \mathbf{z}\ SO_4^{2-} + 8H^+ + I_2$$

What are the balancing numbers **x, y** and **z**?

		x	y	z
☐	**A**	2	1	2
☐	**B**	2	2	4
☐	**C**	2	5	5
☐	**D**	2	5	10

(1 mark)

3 Magnesium reacts with hydrochloric acid.

$$Mg\ (s) + 2HCl\ (aq) \rightarrow MgCl_2\ (aq) + H_2\ (g)$$

Which statement about this reaction is correct?

A. magnesium atoms act as oxidising agents

B. hydrogen molecules act as reducing agents

C. hydrogen ions act as oxidising agents

D. chloride ions act as oxidising agents

(1 mark)

4 The formulae of four ions are shown.

Formula of ion
CrO_4^{2-}
AlO_2^-
$[Fe(CN)_6]^{4-}$
$[CrCl_2(H_2O)_4]^+$

How many of these ions contain a metal with an oxidation number of +3?

A. one

B. two

C. three

D. four

(1 mark)

5 (a) This question is about the reaction shown.

$$2KMnO_4 + xH_2C_2O_4 + yH_2SO_4 \rightarrow 2MnSO_4 + K_2SO_4 + 10CO_2 + zH_2O$$

What values of x, y and z are needed to balance the equation?

		x	y	z
☐	**A**	5	6	8
☐	**B**	10	3	4
☐	**C**	5	3	8
☐	**D**	10	6	4

(1 mark)

(b) What is the reducing agent in the reaction?

A. H^+

B. $C_2O_4^{2-}$

C. MnO_4

D. SO_4

(1 mark)

6 Which name is correct for the ion SO_4^{2-} ?

A. sulfate(II)

B. sulfate(IV)

C. sulfate(VI)

D. sulfate(VIII)

(1 mark)

7 A diprotic acid, H_2A, was titrated with sodium hydroxide solution.

$$H_2A\ (aq) + 2NaOH\ (aq) \rightarrow Na_2A\ (aq) + 2H_2O\ (l)$$

A 25.0 cm^3 portion of 0.100 mol dm^{-3} sodium hydroxide solution required 12.80 cm^3 of the solution of the diprotic acid for complete neutralisation. What is the concentration of H_2A in mol dm^{-3}?

A. 2.56×10^{-2}

B. 9.77×10^{-2}

C. 1.95×10^{-1}

D. 3.91×10^{-1}

(1 mark)

8 In which compound is the oxidation number of nitrogen +5?

A. $Ca(NO_3)_2$

B. Mg_3N_2

C. N_2O_3

D. $NaNO_2$

(1 mark)

9 Which reaction is **not** a redox reaction?

A. $4KClO_3\ (s) \rightarrow 3KClO_4\ (s) + KCl\ (s)$

B. $2HCl\ (aq) + Ba(OH)_2\ (aq) \rightarrow BaCl_2\ (aq) + 2H_2O\ (l)$

C. $Zn\ (s) + CuSO_4\ (aq) \rightarrow ZnSO_4\ (aq) + Cu\ (s)$

D. $Cl_2\ (g) + H_2O\ (l) \rightarrow HCl\ (aq) + HClO\ (aq)$

(1 mark)

10 In which reaction is the copper species acting as an oxidising agent?

A. $Cu^{2+} + 2Ag \rightarrow 2Ag^{+} + Cu$

B. $2Cu^{+} + O^{2-} \rightarrow Cu_2O$

C. $3Cu + O_2 \rightarrow Cu_2O + CuO$

D. $Cu + Hg^{2+} \rightarrow Hg + Cu^{2+}$

(1 mark)

11 Two half-equations for a reaction are shown.

$$Cu \rightarrow Cu^{2+} + 2e^{-}$$

$$NO_3^{-} + 4H^{+} + 3e^{-} \rightarrow NO + 2H_2O$$

What is the overall ionic equation for this reaction?

A. $Cu + NO_3^{-} + 4H^{+} \rightarrow Cu^{2+} + NO + 2H_2O$

B. $2Cu + NO_3^{-} + 4H^{+} \rightarrow 2Cu^{2+} + NO + 2H_2O$

C. $3Cu + 2NO_3^{-} + 8H^{+} \rightarrow 3Cu^{2+} + 2NO + 4H_2O$

D. $6Cu + 2NO_3^{-} + 8H^{+} \rightarrow 6Cu^{2+} + 2NO + 4H_2O$

(1 mark)

12 A titre of 13.25 cm^3 was obtained using a 50 cm^3 burette.

What is the percentage uncertainty in the titre?

[Each reading of the burette has an uncertainty of ±0.05 cm^3]

A. ±0.38%

B. ±0.75%

C. ±1.5%

D. ±7.5%

(1 mark)

13 (a) a)

A pellet of sodium hydroxide has a mass of 0.700 g. Some pellets were dissolved to make 350 cm^3 of 0.25 mol dm^{-3} solution. [M_r value: NaOH = 40] How many pellets were dissolved?

A. 4

B. 5

C. 8

D. 125

(1 mark)

(b) 25.0 cm^3 of the sodium hydroxide solution prepared in (a) was placed in a conical flask and titrated with sulfuric acid.

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

Calculate the number of moles of sulfuric acid that reacted.

A. 0.0031

B. 0.0063

C. 0.013

D. 0.044

(1 mark)

(c) Phenolphthalein indicator was used for the titration in (b).

What was the colour change at the endpoint?

A. colourless → pink

B. pink → colourless

C. orange → yellow

D. yellow → orange

(1 mark)

Hard Questions

1 (a) A group of students carry out an experiment to find the concentration of chlorine, Cl_2 (aq), in a solution.

Excess potassium iodide solution is added to a 10.0 cm^3 sample of the chlorine solution.

$$Cl_2\,(aq) + 2I^-\,(aq) \rightarrow 2Cl^-\,(aq) + I_2\,(aq)$$

The iodine produced is titrated with a solution of thiosulfate ions of known concentration, using starch indicator.

$$2S_2O_3^{2-}\,(aq) + I_2\,(aq) \rightarrow S_4O_6^{2-}\,(aq) + 2I^-\,(aq)$$

The concentration of the Cl_2 (aq) is between 0.038 and 0.042 mol dm^{-3} .

What concentration of thiosulfate ions, in mol dm^{-3} , is required to give a titre of approximately 20 cm^3 ?

A. 0.010

B. 0.020

C. 0.040

D. 0.080

(1 mark)

(b) What is the most suitable volume of 0.1 mol dm^{-3} potassium iodide solution, in cm^3, to add to the 10.0 cm^3 of chlorine solution?

A. 7.6

B. 8.0

C. 8.4

D. 10.0

(1 mark)

(c) What is the colour change at the end-point of the titration?

A. colourless to pale yellow

B. pale yellow to colourless

C. colourless to blue-black

D. blue-black to colourless

(1 mark)